Question			Answer	Marks	AOs		Guidance
10	(a)		$\sin 2x \approx 2x$ or $\sin x \approx x$ used	M1	3.1a	may see $\cos x \approx 1 - \frac{x^2}{2}$	
			$\int \left(\frac{1}{x}\right) dx$ or $\int \left(\frac{1}{x} - x\right) dx$ obtained **oe nfww**	A1	1.1		
			$F[x] = \ln x$ **oe** or $F[x] = \ln x - \frac{1}{2} x^2$ **oe**	A1	1.1	intermediate step needed from here to earn final mark	
			$\ln(0.05) - \ln(0.01) = \ln 5$ **oe** or $\ln(0.05) - \ln(0.01) + 0.0012 \approx \ln 5$ **oe**	A1 **[4]**	3.2a		
10	(b)		differentiation of *their* $\frac{1}{x}$	M1	2.1	or differentiation of y using quotient rule and use of small angle approximation	
			substitution of 0.01 and – 10 000 correctly obtained	A1 **[2]**	1.1	from $-\frac{1}{x^2}$ or $-\frac{1}{x^2} - 1$ **oe**	
10	(c)		4.54066×10^{-6} or 0.00000454066 **cao**	B1	2.5		
			(no sign change for 6 dp), but sign change for 5 dp or last two iterates agree to 5dp	E1	3.1a	allow sign change between 0.947745 and 0.9477475	
			0.94775	B1 **[3]**	3.2a		